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Volume 5, Number 5, September/October 2025

Editorial: Increased Prevalence of Myopia in the United States between 1971 and 1972 and 1999 and 2004—A Reappraisal 100786
Mark A. Bullimore, FCOptom, PhD, Xu Cheng, MD, PhD, Noel A. Brennan, MScOptom, PhD

Original Articles

Retinal Imaging Biomarkers and Correlation to Systemic Disease Activity in Pediatric Sickle Cell Disease 100774
Sandra Hoyek, MD, Celine Chaaya, MD, Colin A. Lemire, BS, Omar Halawa, MD, Francisco Altamirano, MD, Natasha M. Archer, MD, Efren Gonzalez, MD, Nimesh A. Patel, MD

Retinal imaging findings, including OCT and OCT angiography metrics, correlate with systemic disease activity in pediatric sickle cell disease, highlighting associations with hospitalizations, strokes, hemoglobin levels, and transcranial Doppler velocities. This suggests their potential role as systemic biomarkers.

Genotypic Distribution and Clinical Correlations in Familial Exudative Vitreoretinopathy: A Single-Center Study 100782
Brian T. Soetikno, MD PhD, Emily Spoth, MS, M. Elizabeth Hartnett, MD

Wide-angle imaging and next-generation genetic sequencing identified causative variants in 85.7% of families with familial exudative vitreoretinopathy, emphasizing the utility of combining these tools for confirming the clinical diagnosis.

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On the cover: Cover image by Hoang Nguyen, CRA, OCT-C, FOPS and Scott Oliver, MD (UCHealth Sue Anschutz-Rodgers Eye Center, Aurora, CO) was the Second Place winner in the Fundus Photography - Normal category at the Ophthalmic Photographers' Society 2024 Scientific Exhibition (www.opsweb.org).

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Pediatric Eye-Related Emergency Department Visits and Pediatric Eye Provider Location and Density 100776
Julius T. Oatts, MD, Albert Xu, BS, Jonathan R. Morse, BA, John M. Nesemann, MD, Kara M. Cavuoto, MD, Jeremy D. Keenan, MD

In this 10-year population-based study, areas with more pediatric eye providers had fewer eye-related emergency department visits. Each provider was associated with 2.1 fewer visits per 10 000 children. Most eye-related visits were nonurgent.

Is the Incidence of Concomitant Esotropia with Diplopia in the United States Increasing? 100799
Dae Hee Kim, MD, PhD, Scott R. Lambert, MD

Incidence of esotropia with diplopia was analyzed using health insurance claim data from 2007 to 2022. The disease proportion to all strabismus cases tripled from 1.27% to 3.49%. That tendency was prominent among young female patients.

A Multihead Attention Deep Learning Algorithm to Detect Amblyopia Using Fixation Eye Movements 100775
Dipak P. Upadhyaya, MS, Gokce Cakir, MD, Stefano Ramat, PhD, Jeffrey Albert, PhD, Aasef Shaikh, MD, PhD, Satya S. Sahoo, PhD, Fatema Ghasia, MD

This study develops a multihead transformer deep learning model that accurately detects amblyopia using only fixation eye movement data, regardless of amblyopia type, severity, or the presence of nystagmus, demonstrating its potential for clinical application.

Impact of Social Vulnerability Index and Distance to Tertiary Ocular Emergency Department in Patients Presenting with Open Globe Injury 100802
Davina A. Malek, MD, Jason A. Greenfield, BS, Timothy Norris, PhD, Swarup S. Swaminathan, MD, Kara M. Cavuoto, MD

Greater distance from the eye emergency department and higher social vulnerability were associated with worse initial visual acuity in patients presenting with open globe injury but did not significantly impact long-term visual changes or final outcomes.

A Comparative Validation Study of Near Visual Acuity Assessment Using Different Handheld Acuity Charts 100790
David Ziyou Chen, MBBS, FRCOphth, Yih-Chung Tham, PhD, Liang Shen, PhD, Soon-Phaik Chee, MMed (Ophth), FRCOphth

The authors created a conversion table from the Rosenbaum and N-notation charts to the ETDRS near chart. Compared to the ETDRS near chart, the other charts underestimated near vision.

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Corneal Biomechanics as a Causal Factor in Myopia and Astigmatism: Evidence from Mendelian Randomization 100738
Pinghui Wei, PhD, Guoge Han, PhD, Qi Su, MBBS, Lanbo Jia, MBBS, Chao Xue, PhD, Yan Wang, PhD

Mendelian randomization identified corneal hysteresis and resistance factor as protecting against myopia, while corneal-compensated intraocular pressure emerged as a risk factor, highlighting their potential as early biomarkers for predicting and managing refractive errors.

Effect of Myopic Defocus on the Retina and Choroid and Its Interaction with Defocus Regions, Diurnal Rhythm, and Accommodation 100773
Yingying Huang, MD, Jiali Zhang, MD, Xue Li, PhD, Hao Chen, MD, OD, Jinhua Bao, PhD

Sufficient myopic defocus signals in the inferior field, the defocus amounts account for the near-work distance of children, and wearing lenses in the morning and for near work were found to optimize the myopia control effect.

Exploring Effects of Age at the Onset of Myopia on Multiple Diseases Using Electronic Health Records 100819
Xiayin Zhang, PhD, Shan Wang, MD, Yu Huang, PhD, Yanjie Xie, PhD, Ishith Seth, MD, Gabriella Bulloch, MD, Chunran Lai, MD, Yijun Hu, PhD, Xianwen Shang, PhD, Mingguang He, PhD, Zhuoting Zhu, PhD, Honghua Yu, PhD

This study supports effects of age at the onset of myopia on increased risks of retinal detachments, cataracts, primary angle-closure glaucoma, disorders of the vitreous body, and hypothyroidism.

The Association between Myopia and Anxiety: A Bidirectional 2-Sample Mendelian Randomization Study 100832
Waihang Wong, MD, Yimin Qin, MD, Chi Liu, PhD, Shiran Zhang, PhD, Yu Jiang, PhD, Yangfa Zeng, MD, Decai Wang, PhD, Lingyi Liang, PhD, Xiaotong Han, PhD

This study demonstrated a unidirectional causal effect of generalized anxiety disorder on more myopic mean spherical equivalent, indicating the potential benefits of integrated school-based mental health care and myopia interventions at school.

Systematic Review of Proteomics in Age-Related Macular Degeneration and Pathway Analysis of Significant Protein Changes 100793
Olympia Sideri, MD, MSc, Victor Correa, MD, Nikolaos Ziakas, MD, PhD, Ioannis Tsinopoulos, MD, PhD, Joan W. Miller, MD, PhD, Demetrios G. Vavvas, MD, PhD

Unbiased proteomics in age-related macular degeneration (AMD) reveal altered pathways in protease regulation and metabolism, lipid, and complement pathways. They also reveal gaps in data from ocular biofluids in geographic atrophy and early AMD.

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Grip Strength and Age-Related Ocular Diseases: Insights from Observational, Mendelian Randomization, and Mediation Analyses 100831

Chao Chen, Dongling Guo, Jiaqi Meng, MD, Jiao Qi, MD, Keke Zhang, MD, Wenwen He, MD, Yih Chung Tham, PhD, Xiangjia Zhu, PhD

Weaker grip strength is associated with age-related ocular diseases. Mendelian and mediation analyses revealed a genetic link between them, with altered lipids and omega-3 polyunsaturated fatty acids as significant mediators.

Age-Related Scattered Hypofluorescent Spots as an Adverse Prognostic Factor for Polypoidal Choroidal Vasculopathy 100818

Seo Hee Kim, MD, Kai Tzu-iunn Ong, Seonghee Choi, MD, Eun Jee Chung, MD, PhD, Min Kim, MD, PhD, Christopher Seungkyu Lee, MD, PhD, Jinyoung Yeo, PhD, Eun Young Choi, MD, PhD

Using machine learning, age-related scattered hypofluorescent spots on late-phase indocyanine green angiography were identified as a negative prognostic factor in polypoidal choroidal vasculopathy, associated with reduced disease stability and increased recurrence and treatment burden.

Inner Plexiform Layer Substrata Are Discernible with Commercial OCT and Affected by Aging 100815

Victor S.M.C. Correa, MD, Maria Emfietzoglou, MD, Gustavo Sakuno, MD, PhD, Rosanne Naafs, MSc, Joan W. Miller, MD, Alexander Charonis, MD, Demetrios G. Vavvas, MD, PhD

This study examines the sublayers of the inner plexiform layer (IPL) using commercial spectral-domain OCT. Age-related changes to IPL, including thinning and decreased contrast, suggests that it may serve as an early biomarker for synaptic diseases.

Plasma Proteomic Markers of Interleukin-1β Pathway Associated with Incident Age-Related Macular Degeneration in Persons with AIDS 100794

Peter W. Hunt, MD, Adam B. Olshen, PhD, Natalia Murad, PhD, Gabrielle C. Ambayec, BS, Efe Sezgin, PhD, Michael F. Schneider, MS, Douglas A. Jabs, MD, MBA

In a case-control study of patients with AIDS, proteomic evaluation demonstrated a central role for the interleukin-1β pathway in the occurrence of incident intermediate-stage age-related macular degeneration 5 to 10 years later.

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Phase I Study of the Anti-interleukin 33 Fragment Antigen-Binding Region RO7303359 in Geographic Atrophy Secondary to Age-Related Macular Degeneration 100800

Joel A. Pearlman, MD, PhD, Veeral S. Sheth, MD, MBA, Arshad M. Khanani, MD, MA, Vahan B. Indjeian, PhD, Flavia Brunstein, MD, PhD, Denison Kuruvilla, PhD, Mauricio Maia, PhD, Randall Dere, Ling Ma, PhD, Hao Chen, PhD, Seema Datta, PhD, Jeffrey R. Willis, MD, PhD, Henry E. Wiley, MD

A phase I single-dose, dose-escalation study with anti-interleukin 33 Fab in patients with geographic atrophy secondary to age-related macular degeneration demonstrated acceptable safety but failed to demonstrate a discernible effect on aqueous humor biomarkers associated with interleukin 33/ST2 pathway activity.

Oral Antithrombotic Medication Is Associated with Improved Visual Outcomes in Eyes with Submacular Hemorrhage from Wet Age-Related Macular Degeneration 100796

Hemal P. Patel, MD, Cason B. Robbins, MD, Jamie J. Karl, MD, Peter Weng, MD, PhD, Lejla Vajzovic, MD, FASRS, Sharon Fekrat, MD, FASRS

Patients with submacular hemorrhage who are taking oral anthrombotic treatment at the time of onset experience improved visual outcomes compared with those who are not taking any antithrombotic treatment.

Predicting Imminent Conversion to Exudative Age-Related Macular Degeneration Using Multimodal Data and Ensemble Machine Learning 100785

T.Y.Alvin Liu, MD, Yuxuan Liu, MS, Madeleine S. Gastonguay, BS, Dan Midgett, PhD, Nathanael Kuo, PhD, Yujie Zhao, Kareef Ullah, Gwyneth Alexander, Todd Hartman, Neslihan D. Koseoglu, MD, Craig Jones, PhD

Three-dimensional deep learning models, trained with OCT volumes, are capable of predicting both first-eye and fellow-eye imminent conversion to exudative age-related macular degeneration. The addition of clinical data further improved model performance.

Quantification of Hyperreflective Foci in Age-related Macular Degeneration by Polarization-Sensitive OCT 100792

Ryo Terao, MD, PhD, Shuichiro Aoki, MD, PhD, Kohdai Kitamoto, MD, PhD, Kota Totani, AE, Masahiro Yamanari, PhD, Satoshi Sugiyama, MSc, Tatsuya Inoue, MD, PhD, Ryo Obata, MD, PhD, Keiko Azuma, MD, PhD

This study used polarization-sensitive OCT to quantify hyperreflective foci and objectively demonstrated faricimab's efficacy in reducing their amount and area.

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Onset of End-Stage Atrophic Age-Related Macular Degeneration as an End Point—A Delphi Study: Classification of Atrophy Meetings Report 7 100777

Zhichao Wu, BAppSc(Optom), PhD, Srinivas R. Sadda, MD, Thomas Ach, MD, Barbara A. Blodi, MD, Ferdinando Bottoni, MD, Usha Chakravarthy, MD, PhD, Emily Y. Chew, MD, Christine A. Curcio, PhD, Frederick L. Ferris III, MD, Monika Fleckenstein, MD, K. Bailey Freund, MD, Juan E. Grunwald, MD, Frank G. Holz, MD, Glenn J. Jaffe, MD, Sandra Liakopoulos, MD, Tock Han Lim, FRCSEd, Jordi M. Monés, MD, PhD, Sergio Pagliarini, MD, FRCOphth, Daniel Pauleikhoff, MD, Maximilian Pfau, MD, Philip J. Rosenfeld, MD, PhD, David Sarraf, MD, Steffen Schmitz-Valckenberg, MD, Richard F. Spaide, MD, Janet R. Sparrow, PhD, Giovanni Staurenghi, MD, Adnan Tufail, MD, Francesco Viola, MD, Robyn H. Guymer, MBBS, PhD

This study assessed if there was consensus on the acceptability of the onset of endstage atrophic age-related macular degeneration as an endpoint for trials of early interventions, and the criteria for defining such an endpoint.

Pivotal Trial Validating Usability and Visualization Performance of Home OCT in Neovascular Age-Related Macular Degeneration: Report 1 100772

Jeffrey S. Heier, MD, Nancy M. Holekamp, MD, Miguel A. Busquets, MD, Michael J. Elman, MD, Sidney A. Schechet, MD, Byron S. Ladd, MD, Kapil G. Kapoor, MD, Eric W. Schneider, MD, Ella H. Leung, MD, Ron P. Danis, MD, Kester Nahen, PhD, Nishant Mohan, PhD, Gidi Benyamini, MBA

Results of a pivotal study validated the performance of Home OCT system for visualization of hypo-reflective spaces in eyes diagnosed with neovascular age-related macular degeneration toward de novo clearance by the US Food and Drug Administration.

Longitudinal Analysis of Mesopic Microperimetry in a Phase II Trial Evaluating Minocycline for Geographic Atrophy 100783

Thilaka Arunachalam, BS, Maria Abraham, ScM, Christine Orndahl, PhD, Supriya Menezes, MS, Souvick Mukherjee, PhD, Cameron Duic, BS, Minali Prasad, BA, Fares Siddig, BS, Sunil Bellur, MD, Alisa T. Thavikulwat, MD, Clare Bailey, MD, SriniVas R. Sadda, MD, Wai T. Wong, MD, PhD, Emily Y. Chew, MD, Brett G. Jeffrey, PhD, Tiarnán D.L. Keenan, BM BCh, PhD

Longitudinal microperimetry data from a phase II trial of minocycline for geographic atrophy were not consistent with a clinically meaningful effect but demonstrate the advantages and disadvantages of different microperimetry end points for clinical trials.

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Longitudinal Comparison of Geographic Atrophy Enlargement Using Manual, Semiautomated, and Deep Learning Approaches 100787
Jacob Bogost, Rachel E. Linderman, PhD, Robert Slater, PhD, Thomas F. Saunders, OD, Caleb Pacheco, Jeong Pak, PhD, Rick Voland, PhD, Barbara Blodi, MD, Amitha Domalpally, MD, PhD

This study found artificial intelligence is capable of accurately tracking the progression of geographic atrophy in fundus autofluorescence images in about 85% of cases and that human-guided segmentation and artificial intelligence measure geographic atrophy comparably.

Time-in-Range Analysis of Responses after Intravitreal Dexamethasone Therapy in Eyes with Diabetic Macular Edema 100833
Igor Kozak, MD, PhD, Diana V. Do, MD, Hongxin Lai, PhD, Miller J. Ogidigben, PhD, Francisco J. López, MD, PhD

In patients with diabetic macular edema treated with dexamethasone intravitreal implant, evaluation of functional and structural outcomes using a novel time-in-range end point demonstrates longer time with outcomes that correspond to meaningful benefit for patients in clinical settings.

Diabetic Retinopathy Assessment through Multitask Learning Approach on Heterogeneous Fundus Image Datasets 100755
Hongkang Wu, MS, Kai Jin, MD, PhD, Yiyang Jing, MD, Wenyue Shen, MD, Yih Chung Tham, PhD, Xiangji Pan, PhD, Victor Koh, MD, PhD, Andrzej Grzybowski, MD, PhD, Juan Ye, MD, PhD

Diabetic Retinopathy Analysis Model Assistant accurately diagnosed diabetic retinopathy across multisource heterogeneous datasets using multitask learning, with robust performance validated in multicenter studies, significantly outperforming human ophthalmologists, especially in resource-limited settings.

Effect of Faricimab versus Aflibercept on Hyperreflective Foci in Patients with Diabetic Macular Edema from the YOSEMITE/RHINE Trials 100798
Usha Chakravarthy, FRCOphth, PhD, Varun Chaudhary, MD, FRCS(C), Srinivas R. Sadda, MD, Colin S. Tan, FRCOphth, FRCSEd (Ophth), Stela Vujosevic, MD, PhD, Sascha Fauser, MD, Kara Gibson, PhD, Carl Glittenberg, MD, Nancy Holekamp, MD, FASRS, Ben Lanza, PhD, Andreas Maunz, PhD, Jeffrey R. Willis, MD, PhD, Rishi P. Singh, MD

Greater reductions in retinal hyperreflective foci counts and volumes were observed in faricimab-treated eyes compared with aflibercept to week 48 in patients with diabetic macular edema in the phase III YOSEMITE/RHINE trials.

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HORNBILL: A First-in-Human Phase I/IIa Study of the Safety, Tolerability, and Early Pharmacodynamics of BI 764524 for Diabetic Macular Ischemia 100781

Quan Dong Nguyen, MD, MSc, Chirag Jhaveri, MD, Maged Habib, MD, Yasir J. Sepah, MBBS, Khaled Nassar, MD, PhD, Bartlomiej Krawczyk, PhD, Gudrun Simons, PhD, Andrea Giani, MD, Elizabeth Pearce, PhD, Martin Gliem, MD, Mohamed Ahmed, MBBCh, MD, Sobha Sivaprasad, DM, on behalf of the HORNBILL Study Group

HORNBILL was a first-in-human study evaluating the intravitreal anti-Semaphorin 3A antibody BI 764524 in diabetic macular ischemia. All evaluated doses were well tolerated, supporting further investigation of BI 764524 in diabetic retinopathy with retinal nonperfusion.

Developing Laterality-Specific Computable Phenotypes from Electronic Health Record Data, Employing Treatment-Warranted Diabetic Macular Edema as a Use Case 100797

Kaili Ding, MBBS, MS, Tracy Z. Lang, BS, Roberta McKean-Cowdin, PhD, Hossein Ameri, MD, PhD, Narsing A. Rao, MD, Brian C. Toy, MD

Codified clinical datasets have limitations, including missing or erroneous data. An algorithm integrating structured and unstructured data extracted laterality-specific longitudinal information from electronic health records, improving data completeness and accuracy, was validated for diabetic macular edema.

Explainable Artificial Intelligence—Assisted Exploration of Clinically Significant Diabetic Retinal Neurodegeneration on OCT Images 100804

Miyo Yoshida, MD, Tomoaki Murakami, MD, PhD, Kenji Ishihara, MD, PhD, Yuki Mori, MD, PhD, Akitaka Tsujikawa, MD, PhD

Explainable artificial intelligence and expert retinal assessment confirmed 2 known OCT findings and identified 2 novel biomarkers for diabetic neurodegeneration: the ice-pick sign in inner layers and the salt-and-pepper sign in the outer nuclear layer.

Multimodal Imaging and Dark-Adapted Chromatic Perimetry in BEST1 Vitelliform Macular Dystrophy: Identification of Outcome Measurements 100823

Jose Ronaldo Lima de Carvalho, Jr., MD, Jin Kyun Oh, MD, Sara Ragi, BS, Joonpyo Kim, PhD, Remy S. Manzi, MD, Emily Sun, MD, Thiago Cabral, MD, PhD, Rubens Belfort, Jr., MD, PhD, Vivienne C. Greenstein, PhD, Janet R. Sparrow, PhD, Stephen H. Tsang, MD, PhD

Scotopic sensitivity is reduced within the vitelliform lesion of patients with Best vitelliform macular dystrophy and may serve as a valuable outcome measurement for treatment trials of Best disease.

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Computerized Analysis of the Eye Vasculature in a Mass Dataset of Digital Fundus Images: The Example of Age, Sex, and Primary Open-Angle Glaucoma 100778

Jonathan Fhima, MSc, Jan Van Eijgen, MD, PhD, Anat Reiner-Benaim, PhD, Lennert Beeckmans, MSc, Or Abramovich, MSc, Ingeborg Stalmans, MD, PhD, Joachim A. Behar, PhD

This study introduces an automated pipeline for retinal vascular analysis, revealing lower arteriolar and venular diameter, area, tortuosity, and fractal dimension in primary open-angle glaucoma (POAG) and older patients, suggesting pronounced vascular aging in POAG.

Clinical Characteristics and Natural History of Branch Retinal Artery Ischemia 100825

Cortney Connor, BA, Haley S. D'Souza, MD, Raymond Iezzi, MD, MS, Ryan N. Vogel, MD, Ahmad Al-Moujahed, MD, PhD, John B. Miller, MD, Kareem Moussa, MD, Glenn Yiu, MD, PhD

Patients with branch retinal artery ischemia characterized by nonocclusive retinal arterial plaque, sectoral hemorrhages, and delayed arterial flow are associated with systemic vascular risk factors but maintain good vision without complications over many years.

Repeatability of Fractal Analysis in OCT Angiography of the Macula and Optic Nerve Head 100784

Gustavo Rosa Gameiro, MD, PhD, Alessandro A. Jammal, MD, PhD, Verônica Vilasboas-Campos, MD, Jianhua Wang, MD, PhD, Felipe A. Medeiros, MD, PhD

This longitudinal cohort study demonstrated high repeatability of OCT angiography–derived vessel density measurements using fractal analysis and significant correlations to retinal nerve fiber layer thickness, suggesting potential utility for monitoring vascular changes in glaucoma.

Risk of Retinal Vascular Occlusive Disease in Patients with Aortic Stenosis: A Nationwide Korean Cohort Study 100795

Minjeong Kim, MD, Eui-Young Choi, MD, PhD, Kunho Bae, MD, Ju-Yeun Lee, MD, PhD

Aortic stenosis was associated with a twofold increased risk of retinal vascular occlusive diseases over 5 years, even after adjustment for underlying cardiovascular comorbidities.

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Deep Learning–Driven Exophthalmometry through Facial Photographs in Thyroid Eye Disease 100791

Joonhyeon Park, PhD, Jin Sook Yoon, MD, PhD, Namju Kim, MD, PhD, Kyubo Shin, PhD, Hyun Young Park, MD, PhD, Jongchan Kim, MS, Jaemin Park, MS, Jae Hoon Moon, MD, PhD, JaeSang Ko, MD, PhD

The authors developed a deep learning–based system for estimating proptosis in thyroid eye disease using facial photographs that offers an accessible, reliable, and user-friendly tool for objectively monitoring thyroid eye disease.

Next-Generation Sequencing–Based Molecular Profiling of Conjunctival Squamous Cell Carcinoma and Its Potential Application for Therapy 100801

Hakan Demirci, MD, Josh N. Vo, PhD, Yi-Mi Wu, PhD, Victor Elner, MD, Arul M. Chinnaiyan, MD, PhD, Dan Robinson, PhD, F. Yesim Demirci, MD

Invasive conjunctival squamous cell carcinomas are characterized by prevalent human papillomavirus negativity, major roles of altered cell-cycle regulators and chromatin modifiers, and frequent ultraviolet-mediated damage and high tumor mutation burden, with implications for therapy.

Double-Masked, Dose-Response, Vehicle-Controlled Study of VVN461 Ophthalmic Solution in Postoperative Ocular Inflammation 100806

James H. Peace, MD, Kevin Y. Jong, MD, Jason Bacharach, MD, Xiao-Yan Li, MD, Wang Shen, PhD, Caroline Lu, MS, Gary D. Novack, PhD, for the VVN461-CS201 Study Group

In a study of a novel topical ocular Janus kinase inhibitor, VVN461, in subjects undergoing cataract surgery, clinically and statistically significant anti-inflammatory effects were seen.

Diminished MYCN Dosage Endows Cavitary Transformation in Retinoblastoma 100820

Mingpeng Xu, MD, Hanhan Shi, MD, PhD, Yongning Shen, BS, Ludi Yang, MD, Yu Luan, MD, Xiang Gu, MD, Xuyang Wen, MD, PhD, Chuandi Zhou, MD, PhD, Renbing Jia, MD, PhD, Xunda Ji, MD, PhD, Peiquan Zhao, MD, PhD, Minglei Han, MD, PhD, Jiayan Fan, MD, PhD, Peiwei Chai, MD, PhD

This study represents the largest multicentric cohort of cavitary retinoblastoma, unveiling a novel genetic–phenotypic association. Diminished MYCN expression correlates with the formation of translucent cavities and a well-differentiated subtype, with a more favorable prognosis.

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Ultrasonographic Assessment of Intrinsic Vascularity in Choroidal Nevus, Suspicious Choroidal Nevus, and Choroidal Melanoma 100822

Buse Guneri Beser, MD, Tanya McClendon, CDOS, Bernadete Ayres, MD, Hakan Demirci, MD

Some intrinsic vascularity was observed in 86% of non-suspicious choroidal nevi, while all choroidal melanomas demonstrated intrinsic vascularity. Advances in ultrasound technologies and increased experience led to higher detection of intrinsic vascularity in choroidal nevi.



Assessment of Early Fuchs Endothelial Corneal Dystrophy and CTG Trinucleotide Expansion Positivity Using Scheimpflug Imaging 100830

Stanley S.J. Poh, MBBS, Kai Yuan Tey, MBBS, Gary S.L. Peh, PhD, Dawn J.H. Neo, MSc, Hla Myint Htoon, PhD, Evan K.L. Lee, Yu Qiang Soh, MBBS, V Vinod Mootha, MD, Jodhbir S. Mehta, PhD

Fuchs endothelial corneal dystrophy eyes with subclinical edema correlate with increased densitometry and posterior corneal higher-order aberration, indicating early corneal changes, whereas CTG repeat expansion is linked to reduced corneal densitometry, suggesting a distinct disease phenotype.

Scleral Thickness in Eyes with Pachychoroid Pigment Epitheliopathy Accompanied by Keratoconus 100808

Paolo Forte, MD, Alessandro Feo, MD, Hideki Koizumi, MD, PhD, Enrico Borrelli, MD, PhD, Riccardo Manocchio, MD, Luca Di Cello, MD, PhD, Francesco Biagini, MD, Francesco Macocco, MD, Gabriele Drago, MD, Giovanni Forte, MD, Aldo Vagge, MD, PhD, Christian Gianoglio, PhD, Vincenzo Fontana, PhD, Michele Iester, MD, PhD, Massimo Nicolò, MD, PhD, Mario R. Romano, MD, PhD, Chiara Bonzano, MD, PhD

Pachychoroid pigment epitheliopathy affects 1 in 10 patients with keratoconus. Increased scleral thickness emerges as the strongest predictor of this condition, suggesting altered extracellular matrix composition may link anterior and posterior segment pathology in these eyes.

Clinical Efficacy of Ex Vivo Cross-Linking on Neovascularization of the Donor Cornea Carrier Tissue for the Boston Type I Keratoprosthesis 100816

Ana M. Roldan, MD, Rohan Bir Singh, MD, Sofia De Arrigunaga, MD, Elizabeth L. Gatto, Alexander Melki, Steven J. Staffa, MS, David Zurakowski, MS, PhD, Nikolay Boychev, OD, PhD, MEd, Joseph B. Ciolino, MD, for the Keratoprosthesis Cross-linking Study Group

The authors evaluated corneal neovascularization (CNV) in eyes implanted with Boston keratoprosthesis using cadaveric donor tissue, finding a nonsignificant trend toward less neovascularization in cross-linked grafts, suggesting ex vivo corneal cross-linking may reduce CNV risk.

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Regional Variation in Guttae Distribution in Fuchs Endothelial Corneal Dystrophy 100817

Yasufumi Tomioka, MD, PhD, Morio Ueno, MD, PhD, Akihisa Yamamoto, PhD, Koji Kitazawa, MD, PhD, Hideki Fukuoka, MD, PhD, Motomu Tanaka, PhD, Chie Sotozono, MD, PhD, Ula V. Jurkunas, MD, Shigeru Kinoshita, MD, PhD

Slit-scanning widefield contact specular microscopy revealed lower guttae density in the upper eyelid areas of patients with Fuchs endothelial corneal dystrophy, suggesting ultraviolet exposure's role in disease progression and potential preventive strategies.

A Hybrid Deep Learning–Based Approach for Visual Field Test Forecasting 100803

Ashkan Abbasi, PhD, Sowjanya Gowrisankaran, PhD, Wei-Chun Lin, MD, PhD, Xubo Song, PhD, Bhavna Josephine Antony, PhD, Gadi Wollstein, MD, Joel S. Schuman, MD, Hiroshi Ishikawa, MD

Traditional visual field forecasting methods need numerous prior tests, while recent deep learning methods require fewer but face modeling limitations. This study introduces a hybrid deep learning framework to improve flexibility and accuracy in forecasting.

Genome-Wide Association Study of Intraocular Pressure in Population-Based Cohorts in Japan: The Tohoku Medical Megabank Organization Eye Study 100821

Nobuo Fuse, MD, PhD, Hayato Anzawa, PhD, Miyuki Sakurai, PhD, Ikuko N. Motoike, PhD, Satoshi Nagaie, PhD, Tomohiro Nakamura, PhD, Akiko Miyazawa, MD, PhD, Eiichi N. Kodama, MD, PhD, Masatsugu Orui, MD, PhD, Yohei Hamanaka, MD, PhD, Tomoko Kobayashi, MD, PhD, Akira Uruno, MD, PhD, Makiko Taira, MD, PhD, Ritsuko Shimizu, MD, PhD, Naoki Nakaya, PhD, Mami Ishikuro, PhD, Taku Obara, PhD, Fuji Nagami, PhD, Soichi Ogishima, PhD, Fumiki Katsuoka, PhD, Kazuki Kumada, PhD, Shinichi Kuriyama, MD, PhD, Atsushi Hozawa, MD, PhD, Yoko Izumi, MD, PhD, Kengo Kinoshita, PhD, Masayuki Yamamoto, MD, PhD

The authors elucidated the distribution and determinants of intraocular pressure (IOP) in a Japanese population–based genome cohort study. The results revealed genetic similarity and heterogeneity in single-nucleotide polymorphisms related to ethnic variations in IOP.

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A Novel Contact Lens Sensor System for Continuous Intraocular Pressure Monitoring: Evaluation of Accuracy in Human Eyes 100826
Yifan Wei, MM, Yuning Zhang, PhD, Zidong Chen, PhD, Jones Iok-Tong Chong, ME, Christopher Ching Hymn Lee, PhD, Isuru Kaweendra Karunaratne, PhD, Xinyi Zhang, PhD, Mingjie Deng, Yangfan Yang, MD, PhD, David C.C. Lam, PhD, Minbin Yu, MD, PhD

The novel contact lens sensor system provides reliable, continuous intraocular pressure monitoring, showing good agreement with applanation tonometry in seated and supine positions across normal and glaucoma eyes.

Association of Dipeptidyl Peptidase-4 Inhibitors with Glaucoma Risk in Patients with Type 2 Diabetes: A Nationwide Cohort Study 100827
Yi-An Lu, MD, Peng-Tai Tien, MD, PhD, Yih-Dih Cheng, PhD, Yow-Wen Hsieh, PhD, Der-Yang Cho, MD, Shang-Feng Tsai, MD, PhD, Chien-Hsiang Weng, MD, MPH, Heng-Jun Lin, MS, Hui-Ju Lin, MD, PhD, I-Jong Wang, MD, PhD, Chien-Chih Chou, MD, PhD

Utilizing Taiwan's national insurance database, this study demonstrates that the use of dipeptidyl peptidase-4 inhibitors is significantly associated with reduced risks of primary open-angle glaucoma and normal-tension glaucoma in patients with type 2 diabetes.

Correspondence (available at www.ophthalmologyscience.org/current)

Re: Chaurasia et al: Highly Accurate and Precise Automated Cup-to-Disc Ratio Quantification for Glaucoma Screening
Eunice Jin Hui Goh, MMed (Ophth), FRCOphth, Boon Peng Yap, BEng, Kelvin Zhenghao Li, MMed (Ophth), FRCOphth

Reply
Abadh K. Chaurasia, PhD, Connor J. Greatbatch, MBBS, Xikun Han, PhD, Puya Gharahkhani, PhD, David A. Mackey, MD, FRANZCO, Stuart MacGregor, PhD, Jamie E. Craig, PhD, FRANZCO, Alex W. Hewitt, MBBS, FRANZCO

Myopia

Light Inhibits Lens-Induced Myopia through an Intensity-Dependent Dopaminergic Mechanism 100779
Cindy Karouta, PhD, Kate Thomson, PhD, Ian Morgan, PhD, Regan Ashby, PhD

As observed for form-deprivation myopia (FDM), bright light inhibits lens-induced myopia (LIM) in an intensity-dependent manner through a dopaminergic mechanism. However, unlike FDM, light could not abolish LIM due to an active defocus cue.

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Ophthalmology® Science

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Clinical Features and Natural Progression of Unilateral High Myopia in Adults: A Comparison Study 100780

Dong Geun Kim, MD, Seok Hyun Bae, MD, Dong Ju Kim, MD, Jong Suk Lee, MD, Kwangsic Joo, MD, PhD, Sang Jun Park, MD, PhD, Se Joon Woo, MD, PhD, Kyu Hyung Park, MD, PhD

This study compares unilateral and bilateral high myopia in adults, showing that unilateral high myopia eyes have worse vision despite shorter axial length, while the fellow eye remains stable. Further research into genetic factors is recommended.

The Noninvasive Measurement of Ocular Rigidity and the Pulsatile Choroidal Volume Change in Children and Young Adults 100814

Martin Reyes, Sage Smith, Kelly Lam, Yasaman Adel, Nimesh B. Patel, OD, PhD, Diane N. Sayah, OD, PhD

Measures of ocular rigidity and pulsatile choroidal volume change are achievable in children and have excellent repeatability. This noninvasive method can be used to establish structural biomechanical changes of the sclera and choroid in myopia.

Long-Shaft Vitrectomy Probe for Vitreoretinal Diseases in Highly Myopic Eyes: A Randomized Controlled Trial 100824

Yun Hsia, MD, MS, Cheng-Yung Lee, MD, Mei-Chi Tsui, MD, Shih-Wen Wang, MD, Chien-Jung Huang, MD, I-Hsin Ma, MD, MS, Kuo-Chi Hung, MD, Muh-Shy Chen, MD, PhD, Zih-Wei Yang, MD, Bo-Da Huang, MD, Ting-Chieh Ko, MD, Tzyy-Chang Ho, MD

The long-shaft vitrectomy probe improves posterior pole access for internal limiting membrane trimming and cortical vitreous removal in highly myopic eyes, especially those with axial length >28 mm. Addressing instrument bending could further optimize its design.

The journal's Guide for Authors and Guide for Reviewers are available at www.ophthalmologyscience.org.